

STAT 9120 Midterm Review Sheet

Theory of Linear Models - Fall 2026

Estimability, model updating, multivariate normal and chi-squared distributions, quadratic forms, Cochran's theorem, and hypothesis testing

Big picture. Most of the midterm can be organized around four ideas:

1. **Linear algebra determines what is identifiable/testable:** column spaces, null spaces, rank, projections.
2. **Least squares is projection:** fitted values live in $\mathcal{C}(X)$ and residuals live in $\mathcal{C}(X)^\perp$.
3. **Normality plus orthogonal geometry gives exact distributions:** projections of Gaussian vectors lead to chi-squared variables, and orthogonal projections give independence.
4. **Hypothesis tests compare model spaces:** the numerator of an F statistic measures squared length in directions allowed by the full model but excluded by the null model.

1. Core linear-model geometry to know

Assume, unless otherwise stated,

$$Y = X\beta + \varepsilon, \quad E(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2 I_n,$$

with $X \in \mathbb{R}^{n \times p}$. If X has full column rank,

$$\hat{\beta} = (X^T X)^{-1} X^T Y, \quad H = X(X^T X)^{-1} X^T, \quad M = I - H.$$

Then

$$\hat{Y} = HY, \quad e = MY,$$

and

$$H^T = H, \quad H^2 = H, \quad M^T = M, \quad M^2 = M, \quad HM = 0.$$

Important subspaces and dimensions:

$$\mathcal{C}(H) = \mathcal{C}(X), \quad \mathcal{C}(M) = \mathcal{C}(X)^\perp = \mathcal{N}(X^T),$$

$$\text{rank}(H) = p, \quad \text{rank}(M) = n - p.$$

Also remember

$$\mathcal{N}(X^T X) = \mathcal{N}(X), \quad \text{rank}(X^T X) = \text{rank}(X).$$

2. Estimability

A linear parametric function $c^T \beta$ is **estimable** if there exists a such that

$$E(a^T Y) = c^T \beta \quad \text{for every } \beta.$$

Since $E(a^T Y) = a^T X\beta = (X^T a)^T \beta$, the defining condition is

$$c^T \beta \text{ estimable} \iff c \in \mathcal{C}(X^T).$$

Equivalent characterizations:

$$c \in \mathcal{C}(X^T) \iff c \in \mathcal{N}(X)^\perp \iff c^T h = 0 \quad \forall h \in \mathcal{N}(X).$$

Interpretation. If $X\beta_1 = X\beta_2$, then $h = \beta_1 - \beta_2 \in \mathcal{N}(X)$. An estimable function must satisfy

$$c^T \beta_1 = c^T \beta_2.$$

Thus an estimable function is constant across all parameter vectors that generate the same mean vector.

Rank-deficient least squares

If $X^T X$ is singular and G is a generalized inverse satisfying

$$(X^T X)G(X^T X) = X^T X,$$

one least-squares solution is

$$\hat{\beta}_G = GX^T Y.$$

The vector $\hat{\beta}_G$ may depend on G , but

$$X\hat{\beta}_G$$

and every estimable quantity

$$c^T \hat{\beta}_G$$

are unique.

Common exam mistake: nonuniqueness of β does *not* imply nonuniqueness of the fitted mean $X\beta$. Always separate the parameterization from the model space $\mathcal{C}(X)$.

3. Model updating

Adding one covariate

Suppose the current model has design matrix X and a new covariate z is added. Let

$$M_X = I - H_X, \quad z^* = M_X z.$$

The vector z^* is the part of z orthogonal to the current model space. If $z^* \neq 0$, then

$$H_+ = H_X + \frac{z^* z^{*T}}{z^{*T} z^*}.$$

The additional term is a rank-1 projection onto the new direction contributed by z .

The reduction in residual sum of squares is

$$\text{SSE}_X - \text{SSE}_+ = \frac{(z^{*T} Y)^2}{z^{*T} z^*}.$$

The coefficient of z in the enlarged model is

$$\hat{\gamma} = \frac{z^T M_X Y}{z^T M_X z}.$$

This is the Frisch-Waugh-Lovell result: residualize both Y and z against X , then regress residual on residual.

If $z^* = 0$, then $z \in \mathcal{C}(X)$, so the new variable adds no new model-space direction.

Adding one observation

Let

$$A = (X^T X)^{-1}.$$

For a new predictor vector x_0 , the updated cross-product matrix is

$$X_+^T X_+ = X^T X + x_0 x_0^T.$$

Sherman-Morrison gives

$$(X_+^T X_+)^{-1} = A - \frac{A x_0 x_0^T A}{1 + x_0^T A x_0}.$$

The scalar

$$h_0 = x_0^T A x_0$$

measures how far the new design point lies in a direction that is weakly represented by the existing design.

Deleting one observation

For observation i with residual e_i and leverage h_{ii} ,

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{(X^T X)^{-1} x_i e_i}{1 - h_{ii}}.$$

Thus large influence requires some combination of

- a large residual $|e_i|$, and/or
- large leverage h_{ii} .

The deleted residual identity is

$$e_{(i)} = \frac{e_i}{1 - h_{ii}}.$$

This identity is also the basis of leave-one-out prediction calculations.

4. Multivariate normal distribution

If

$$Y \sim N_n(\mu, \Sigma),$$

then for any fixed matrix A and vector b ,

$$AY + b \sim N(A\mu + b, A\Sigma A^T).$$

Special cases:

- $a^T Y \sim N(a^T \mu, a^T \Sigma a)$.
- Subvectors of an MVN vector are MVN.
- Any collection of linear combinations is jointly normal.

For jointly normal vectors U and V ,

$$\text{Cov}(U, V) = 0 \implies U \perp V.$$

This implication is generally false outside the multivariate normal family.

For the normal linear model,

$$Y \sim N_n(X\beta, \sigma^2 I),$$

so

$$\hat{\beta} \sim N_p(\beta, \sigma^2 (X^T X)^{-1})$$

when X has full column rank.

Also,

$$e = MY \sim N_n(0, \sigma^2 M).$$

Although e has n coordinates, they are not independent because M is singular with rank $n - p$.

Independence of coefficients and residuals.

$$\text{Cov}(\hat{\beta}, e) = \sigma^2 (X^T X)^{-1} X^T M = 0.$$

Because $\hat{\beta}$ and e are jointly normal,

$$\hat{\beta} \perp e.$$

Consequently, $\hat{\beta}$ is also independent of $\text{SSE} = e^T e$.

5. Chi-squared distributions

If

$$Z_1, \dots, Z_r \stackrel{\text{iid}}{\sim} N(0, 1),$$

then

$$\sum_{j=1}^r Z_j^2 \sim \chi_r^2.$$

Useful moments:

$$E(\chi_r^2) = r, \quad \text{Var}(\chi_r^2) = 2r.$$

The degrees of freedom count independent squared standard-normal directions.

For a shifted normal vector, squared lengths lead to a noncentral chi-squared distribution. If

$$U \sim N_r(\delta, I_r),$$

then

$$U^T U \sim \chi_r^2(\lambda), \quad \lambda = \delta^T \delta.$$

6. Quadratic forms

A quadratic form is

$$Q = Y^T A Y.$$

Only the symmetric part of A matters because

$$Y^T A Y = Y^T \left(\frac{A + A^T}{2} \right) Y.$$

If A is symmetric, write its spectral decomposition as

$$A = Q \Lambda Q^T,$$

where Q is orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. For $Z \sim N(0, I)$,

$$Z^T A Z \stackrel{d}{=} \sum_{j=1}^n \lambda_j U_j^2,$$

where $U_j \stackrel{\text{iid}}{\sim} N(0, 1)$.

Therefore a general symmetric quadratic form is a weighted sum of independent χ_1^2 variables.

The key projection result

If P is symmetric and idempotent,

$$P = P^T = P^2,$$

then its eigenvalues are only 0 and 1. If $r = \text{rank}(P)$,

$$Z^T P Z \sim \chi_r^2.$$

Geometrically,

$$Z^T P Z = \|PZ\|^2,$$

the squared length of the Gaussian vector projected onto an r -dimensional subspace.

If

$$Y \sim N(\mu, \sigma^2 I),$$

then

$$\frac{Y^T P Y}{\sigma^2} \sim \chi_r^2(\lambda), \quad \lambda = \frac{\mu^T P \mu}{\sigma^2}.$$

The form is central exactly when

$$P\mu = 0.$$

Useful moments

For $Y \sim N(\mu, \Sigma)$ and symmetric A ,

$$E(Y^T A Y) = \text{tr}(A \Sigma) + \mu^T A \mu,$$

$$\text{Var}(Y^T A Y) = 2 \text{tr}(A \Sigma A \Sigma) + 4 \mu^T A \Sigma A \mu.$$

7. Cochran's theorem

A useful projection-matrix version is:

Suppose $P_1, \dots, P_k$ are symmetric idempotent matrices such that

$$P_i P_j = 0 \quad (i \neq j), \quad \sum_{j=1}^k P_j = I.$$

If $Z \sim N_n(0, I)$ and $r_j = \text{rank}(P_j)$, then

$$Z^T P_j Z \sim \chi_{r_j}^2 \text{ independently.}$$

The rank identity

$$r_1 + \dots + r_k = n$$

expresses a decomposition of $\mathbb{R}^n$ into mutually orthogonal subspaces.

Do not infer independence merely from an additive decomposition. You need the relevant orthogonality conditions (or an equivalent Cochran-theorem argument).

8. Cochran's theorem in regression

Assume the model contains an intercept and define

$$J = \frac{1}{n} \mathbf{1} \mathbf{1}^T.$$

Then

$$I - J = (H - J) + (I - H),$$

with

$$(H - J)(I - H) = 0.$$

The corresponding sums of squares are

$$\text{SSE} = Y^T (I - H) Y,$$

$$\text{SSR} = Y^T (H - J) Y,$$

$$\text{SST} = Y^T (I - J) Y.$$

Their ranks are

$$\text{rank}(I - H) = n - p, \quad \text{rank}(H - J) = p - 1, \quad \text{rank}(I - J) = n - 1.$$

Under the global null (all slopes zero),

$$\frac{\text{SSR}}{\sigma^2} \sim \chi_{p-1}^2, \quad \frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p}^2, \quad \text{SSR} \perp \text{SSE}.$$

Thus

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} \sim F_{p-1, n-p}.$$

Also,

$$\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p}^2$$

for every β , because $(I - H)X\beta = 0$. Hence

$$E(\text{MSE}) = \sigma^2, \quad \text{MSE} = \frac{\text{SSE}}{n-p}.$$

9. Hypothesis testing in linear models

General linear hypothesis

For a full-rank model, consider

$$H_0 : C\beta = d,$$

where C is $q \times p$ with rank q . Under H_0 ,

$$C\hat{\beta} - d \sim N_q(0, \sigma^2 C(X^T X)^{-1} C^T).$$

Therefore

$$Q = (C\hat{\beta} - d)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d)$$

satisfies

$$\frac{Q}{\sigma^2} \sim \chi_q^2.$$

Replacing σ^2 with independent MSE gives

$$F = \frac{Q/q}{\text{MSE}} \sim F_{q, n-p} \quad \text{under } H_0.$$

For $q = 1$,

$$t = \frac{c^T \hat{\beta} - d}{\sqrt{\text{MSE } c^T (X^T X)^{-1} c}} \sim t_{n-p},$$

and

$$F = t^2.$$

Nested-model formulation

If

$$\mathcal{C}(X_0) \subset \mathcal{C}(X_1)$$

with ranks $p_0 < p_1$, then

$$H_1 - H_0$$

is the projection onto the additional directions supplied by the full model, with rank

$$q = p_1 - p_0.$$

Moreover,

$$\text{SSE}_0 - \text{SSE}_1 = Y^T (H_1 - H_0) Y.$$

Under the reduced-model null,

$$F = \frac{(\text{SSE}_0 - \text{SSE}_1)/q}{\text{SSE}_1/(n - p_1)} \sim F_{q, n-p_1}.$$

Geometry of the partial F test. The numerator measures the squared length of the data in the q directions present in the full model but absent from the reduced model. The denominator measures residual squared length in the orthogonal complement of the full model. Cochran's theorem supplies independence.

Rank-deficient models

In a rank-deficient model, a hypothesis

$$H_0 : C\beta = d$$

is meaningful only if the rows of C represent estimable functions. For a single row c^T ,

$$c^T \beta \text{ is testable} \iff c \in \mathcal{C}(X^T).$$

Identifying constraints can change the numerical parameterization, but they cannot change the value or test result for an estimable function.

10. One-page formula map

Concept	Formula / criterion
Estimability	$c^T \beta$ estimable $\iff c \in \mathcal{C}(X^T) \iff c \perp \mathcal{N}(X)$
Projection	$H = X(X^T X)^{-1} X^T$, $M = I - H$
Add covariate	$z^* = M_X z$, $H_+ = H_X + z^* z^{*T} / (z^{*T} z^*)$
FWL	$\hat{\gamma} = (z^T M_X Y) / (z^T M_X z)$
Delete observation	$\hat{\beta}_{(i)} = \hat{\beta} - (X^T X)^{-1} x_i e_i / (1 - h_{ii})$
MVN transform	$Y \sim N(\mu, \Sigma) \Rightarrow AY + b \sim N(A\mu + b, A\Sigma A^T)$
Joint normal independence	$\text{Cov}(U, V) = 0 \Rightarrow U \perp V$
Chi-squared	$\sum_{j=1}^r Z_j^2 \sim \chi_r^2$
Projection quadratic form	$P = P^T = P^2$, $\text{rank}(P) = r \Rightarrow Z^T P Z \sim \chi_r^2$
Noncentrality	$Y^T P Y / \sigma^2 \sim \chi_r^2(\mu^T P \mu / \sigma^2)$
Residual SSE	$\text{SSE} / \sigma^2 \sim \chi_{n-p}^2$
General F test	$F = \{Q/q\} / \text{MSE} \sim F_{q, n-p}$ under H_0
Nested F test	$F = \{(\text{SSE}_0 - \text{SSE}_1)/q\} / \{\text{SSE}_1/(n - p_1)\}$
Single restriction	$F = t^2$

11. Common mistakes to avoid

- Confusing $\mathcal{C}(X)$ with $\mathcal{C}(X^T)$.
- Assuming every component of β is estimable.
- Forgetting that generalized-inverse solutions can differ even when fitted values do not.
- Forgetting to residualize a new covariate against the current design before interpreting its added direction.
- Treating uncorrelated variables as independent without joint normality.
- Assuming any quadratic form in normals is chi-squared.
- Forgetting that chi-squared projection results require a symmetric idempotent matrix.
- Forgetting the mean condition for a *central* chi-squared form: $P\mu = 0$.
- Claiming independence just because two sums of squares add to a total.
- Memorizing degrees of freedom without relating them to ranks.
- Using $n - p_0$ instead of $n - p_1$ in the denominator of a partial F test.
- Testing a nonestimable function in a rank-deficient model.

12. Self-check problems

These are intended as short review prompts; you should be able to answer each without extensive computation.

1. If $h \in \mathcal{N}(X)$ and $c^T h \neq 0$, what does that tell you about $c^T \beta$?
2. A new covariate satisfies $M_X z = 0$. What happens to the fitted values and SSE after adding it?
3. If $Y \sim N(\mu, \Sigma)$, what are the mean and covariance of $AY + b$?
4. Let P be a rank-4 orthogonal projection and $Z \sim N(0, I)$. What is the distribution of $Z^T P Z$?
5. If $Y \sim N(\mu, \sigma^2 I)$ and $P\mu \neq 0$, what changes about the quadratic-form distribution?
6. In a full-rank model with $n = 40$ and $p = 6$, what is the distribution of SSE / σ^2 ?
7. For nested models with $p_0 = 3$ and $p_1 = 7$, how many numerator degrees of freedom does the partial F test have?
8. Why are the extra sum of squares and full-model SSE independent under the null?
9. What is the relationship between the two-sided t test of one restriction and its corresponding F test?
10. In a rank-deficient model, how do you determine whether a proposed linear hypothesis is testable?